

Holding out whole tokamaks reverses machine-learning rankings for energy-confinement scaling

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Confinement scaling laws are fitted to existing tokamaks and used to size ones not yet built, so what matters is accuracy on an unseen device. On the ITPA Global H-mode Confinement Database (6228 time slices, 16 devices), cross-validation that holds out discharges ranks a random forest 29% more accurate than a refitted power law. Holding out whole devices reverses the ranking: the forest is worse than the power law on all 11 devices with enough data to score. The reversal survives, though not on every device, when stored energy replaces confinement time as the target and when the features are replaced by dimensionless groups that carry less device identity. It reproduces on all 7 eligible devices of the stellarator-heliotron confinement database, in an analysis fixed before those data were downloaded. At a size cut matching ITER's 1.82-fold step in major radius, 90% split-conformal intervals cover 3% and 0% of held-out rows for the two tree ensembles, with widths that change by under 1.5%, so they give no sign of failing. Connor–Taylor constraints on the exponents and a Gaussian process with a linear kernel term both improve extrapolation. An ablation that varies only the long-range mean points to saturation, rather than flexibility alone, as the cause in the models tested. Four reporting checks follow, three of which need no extra fitting.

Keywords: tokamak energy confinement; confinement scaling; machine learning; extrapolation; cross-validation; uncertainty quantification; ITER

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I. INTRODUCTION

A tokamak ignites when the alpha heating from its own fusion reactions covers everything the plasma loses. The energy confinement time τ_E sets that loss rate, so it is a principal input to how large a next-step device has to be. Empirical scaling remains a principal tool for projecting it to a device that does not exist yet¹: a power law in the engineering parameters,

$$\tau_E = C I_p^{a_1} B_t^{a_2} \bar{n}_e^{a_3} P^{a_4} R^{a_5} \epsilon^{a_6} \kappa^{a_7} M^{a_8}, \quad (1)$$

fitted by least squares across discharges from the multi-machine confinement database. In log space Eq. (1) is a linear model. The reference fit is IPB98(y,2)¹, whose exponents were used to size ITER. The database is still maintained and republished², and the 2007 revision of the ITER physics basis kept IPB98(y,2) as the reference H-mode scaling³.

What such a law is used for is an extrapolation, and a large one. ITER’s major radius is 1.82 times that of the largest machine in the database, and SPARC⁴ is designed for a toroidal field 2.1 times the largest recorded in it. First-principles and reduced transport models offer a second route to the same number, and the machine learning in that branch trains neural surrogates to reproduce a physics model fast enough to run inside a simulation^{5,6}; SPARC’s core performance is projected that way⁷ alongside the empirical route. Nothing below concerns that branch. The empirical law is still fitted, published, quoted as a design point and used to benchmark the physics-based projections, and this paper is about how empirical models are chosen.

Replacing Eq. (1) with a flexible regressor is an old idea. Allen and Bishop⁸ fitted a neural network to confinement data in 1992 and found that it beat linear regression on an in-distribution score. Machine learning has since been productive in disruption prediction^{9–11} and in magnetic control¹², and confinement scaling has been revisited with symbolic regression¹³, Bayesian neural networks¹⁴, tuned tree ensembles¹⁵, deep ensembles with features aligned across machines¹⁶, and a Kolmogorov–Arnold network learned as a residual on a power law¹⁷. Measured the usual way, the flexible models win. On the rows used here, a random forest scores 0.128 in log-RMSE against 0.181 for a power law refitted in the same folds, a margin of 29%, and 0.199 for the published IPB98(y,2) law, a margin of 36%.

That comparison holds out discharges, and every machine in a held-out fold also appears in the training folds. A held-out row has hundreds of near neighbours recorded on the

same device at nearby settings, so the score measures how much of JET is predictable from the rest of JET. A scaling law is used on a machine that is not in the database at all. Fields whose data arrive in clusters match the validation split to the generalisation being claimed¹⁸: materials informatics adopted leave-one-cluster-out cross-validation to measure extrapolation to unseen chemistries¹⁹, and treats distance from the training data as the applicability domain of a model²⁰. Inside fusion, Abbate *et al.*²¹ make the same argument for transport models: score them on a large database, against a baseline chosen for the use they are put to.

This paper applies that principle to confinement scaling on the International Tokamak Physics Activity (ITPA) database, and reports four results.

1. *The ranking of the models reverses.* Held out one device at a time, the random forest that wins cross-validation is worse than the refitted power law on all 13 eligible database labels and all 11 eligible physical devices. Between a power law and the two tree ensembles, the usual procedure therefore does more than overstate accuracy: it selects the worst extrapolator. The same comparison, run under rules fixed and archived before the data were downloaded on the stellarator-heliotron confinement database²², which shares no device with the tokamak one, reproduces the inversion on every eligible held-out device.
2. *Calibrated intervals fail without warning.* At a size cut matched to ITER’s step in major radius, split-conformal intervals set to 90% cover 3% and 0% of held-out rows for the two tree ensembles, and their width moves by under 1.5%.
3. *Long-range structure improves extrapolation.* Connor–Taylor similarity constraints on the exponents and a Gaussian process with a linear kernel term both extrapolate well, and an ablation that changes only the long-range mean points to saturation, rather than flexibility alone, as the cause in the models tested.
4. *A short reporting convention follows,* and three of its four items need no refit.

The closest prior work covers parts of this picture. Wang *et al.*¹⁷, on the same database revision, find that an unanchored network behaves unpredictably outside the training distribution and that anchoring it to a power law helps. Their held-out cohorts are defined by thresholds on parameter ranges, inside the population of machines already seen, and

they note that their approach cannot substitute for device coverage. Butt *et al.*²³ hold out whole machines on the same standard set and find that adding parameters to a power law improves the in-sample fit and worsens extrapolation, but they score no flexible model. Nam and Seo¹⁶ report that transfer across machines is harder than within them. This paper holds out the device and compares model families under that split, so what it finds is an inversion between a power law and flexible models, where Butt *et al.* find a loss of accuracy inside one family. It also measures what the same split does to a calibrated interval, and it separates flexibility from long-range saturation, which tells a modeller what property to avoid without singling out a best model.

II. DATA AND METHODS

a. Data. The ITPA Global H-mode Confinement Database, standard analysis set STD5 of revision DB5.2.3², obtained from the Open Science Framework (OSF) deposit²⁴. The file holds 6228 rows from 4471 discharges across 18 device/wall labels, which are 16 physical devices once the two JET entries and the two ASDEX Upgrade entries are each merged. No row fails the cleaning rule, which requires every used column to be finite and positive, and no outlier rule, winsorising or reweighting is applied. The database description gives STD5 as 7537 points from 18 machines; the deposit delivers 6228 under that name, and they are analysed as delivered. This study did not establish where the difference comes from. The target is the thermal confinement time **TAUTH**. The file is pinned by SHA-256 and verified on load, so every number below refers to one set of bytes.

b. Features and models. Nine log engineering features: plasma current, toroidal field, line-averaged density, loss power, major radius, elongation, inverse aspect ratio, effective mass and minor radius. The candidates are a ridge regression on the log features, which is a fitted power law; a random forest of 300 trees²⁵ in which every split considers all nine features; a histogram gradient booster; log-quadratic and log-cubic ridge controls; and a mean baseline. The power law and the two tree ensembles are the primary comparison. Beside the mean baseline, they were the only candidates in the first committed version of the pipeline, which predates the first held-out-device analysis in the repository's history. The polynomial and Gaussian-process models joined with or after that analysis, as controls. None is tuned for the headline comparison. Every consequential setting is written out in

Sec. S7 of the supplementary material, and Sec. IIIB reports a nested search. Minor radius is reconstructed as $a = \epsilon R$, so the nine features carry one exact dependency; the refit of Eq. (1) on the eight independent variables plus an intercept has condition number 10.7, and Sec. S14 audits the rank of the design and the refitted exponents. IPB98(y,2), evaluated analytically, is a historical reference and not a competitor. It was fitted in 1998 to DB2.8, a predecessor that overlaps this database heavily in devices and regimes², so it has seen most of the machines held out below. Its prediction is never used as a feature, and the like-for-like baseline throughout is the power law refitted inside each fold.

c. Splits. (i) Cross-validation grouped by discharge: five folds of **GroupKFold** on the discharge identifier, so every slice of one discharge falls in the same fold. (ii) Holding out a whole unit, in two versions. Leave-one-label-out (LOLO) holds out each of the 13 database labels with at least 30 rows in turn and trains on every other row. Leave-one-device-out (LODO) holds out each of the 11 physical devices those labels reduce to. Holding out JET-ILW while JET stays in training does not hold out an unseen device, so every claim that turns on an unseen device is made on LODO. Merging the wall eras has a cost of its own, since the ITER-like wall brought a documented confinement reduction concentrated in the pedestal²⁶. The 30-row threshold was fixed before any held-out score was computed and decides only which labels are scored: the five smaller labels (COMPASS, TCV, START, TDEV and TFTR, 43 rows between them) stay in every training fold. (iii) A size-ordered cut, described in Sec. IV.

d. Scores and statistics. Every model is fitted and scored in $\log \tau$, and the score is the root-mean-square residual in that variable, written log-RMSE. Intervals are percentile bootstraps over whole held-out units, 2000 resamples over the 13 labels; paired intervals resample the per-label difference between two models, which removes the difficulty the labels share. The folds of a leave-one-out arm share almost all of their training rows, so the units they score are not independent experiments. The bootstrap intervals and the sign-test and permutation p -values below summarise spread and how lopsided a count is. The claims rest on the win counts and the size of the gaps.

e. External replication. One external replication was specified before any of its scores were seen. Its rules, the population, the feature mapping, the eligibility threshold and a five-way verdict rule that includes “cannot be evaluated” and “contradicts the original result” were committed to the repository and archived before the data file was downloaded, and the

analysis was run once. The dataset is the International Stellarator-Heliotron Confinement Database, the database the ISS04 scaling²² was fitted to. A stellarator has no plasma current, so the regressors are ISS04’s rather than the nine above, and the comparison between the two databases is of the pattern and not of any number.

f. Software. Python 3.12.14, scikit-learn 1.7.2, NumPy 2.2.6, SciPy 1.18.1 and pandas 2.3.3, each pinned in the repository’s `constraints.txt`. Every estimator is seeded, and reruns are bit-identical.

Use of generative AI. Generative AI tools were used in this work, and each is named here with its maker, its version and its role. Anthropic’s Claude models, used through Anthropic’s Claude Code tool, assisted with developing, debugging and executing author-specified analysis code in the accompanying repository, and with drafting and editing passages of this manuscript: Claude Opus 5 (model `claude-opus-5`) until mid-September 2026, and Claude Fable 5.1 (model `claude-fable-5-1`) from then on, including the external replication of Sec. S17 of the supplementary material and the revision and shortening of the manuscript in September 2026. One restructuring of the repository, in July 2026, used Claude Opus 4.8. The repository’s commit history records which model assisted with each change. OpenAI’s GPT-5.6 Sol, in August and September 2026, was used to critique complete drafts against journal policy and for scientific and historical accuracy, to support the literature review, and to identify methodological issues and propose robustness checks. Several of its suggestions were adopted after the author judged them worth pursuing, among them the corrected account of how IPB98(y,2) was fitted (Sec. VI A), the recomputation of the distance correlation over the 11 physical devices (Sec. III A), the device-level calibration (Sec. V), and several of the references in Secs. S3 and S14 of the supplementary material, which the author then located and checked against their published records. The author, a sole researcher without funding, used these tools because they made it practical to write, test and document analysis code and a manuscript of this scope. None of these systems is an author, and none supplied any data, result, figure or reference directly: every number and figure is produced by the repository’s scripts from the pinned inputs, and every reference was located by the author and checked against its published record. The author designed the study, decided which suggestions to pursue, specified and implemented every analysis reported here, verified every reported number against the generated artefacts, and takes full responsibility for the content, the conclusions and any error in them. The drafts are preserved in the repository’s

history, and the author retains the transcripts of the sessions.

III. HOLDING OUT A DEVICE REVERSES THE RANKING

Table I is the central result. Of the three primary models, a power law, a random forest and a gradient booster, the forest is best under cross-validation and worst on a label it has not seen. Against the power law refitted in the same folds it is 29% better in the CV column and 117% worse in the LOLO column. What holds in every partition tested is that the power law and the two ensembles trade places. The order of the two ensembles between themselves does not hold: they are 0.002 apart under cross-validation, and over ten shuffled partitions the forest is ahead of the booster in 2 of 10. The two polynomial rungs are controls, outside the ranking claim.

The labels differ enormously in difficulty, so the right test is the paired per-label difference. The random forest is worse than the log-linear power law on **13 of 13** database labels, at a mean gap of +0.251 with a 95% interval of [+0.157, +0.342], and the gradient booster is worse on 12 of 13. Over the 11 physical devices the forest loses on all 11, and the gap widens to +0.315, [+0.221, +0.408]. The inversion is a statement about these three models. The lowest cross-validated score in this study, 0.112, belongs to the Gaussian process of Sec. VIB, and it does not reverse: on a held-out label it scores 0.218 against the power law’s 0.214, a tie. Figure 1 shows the reversal with the diagnostics of the next subsection.

A. Why the forest fails

a. Distance. Ranking each model’s per-label error against the Mahalanobis distance of that label’s mean log-feature vector from the training mean gives $\rho = +0.85$ for the random forest (permutation $p = 0.0007$) and +0.54 for the gradient booster, against $\rho = -0.06$ for the log-linear power law. Over the 11 physical devices the forest’s correlation is +0.77 and the power law’s is still -0.06 . The mean baseline, which fits nothing, also correlates with distance, at +0.56. So distance measures how hard a unit is in general; the difference is that the power law’s error does not grow with it. Mahalanobis distance between mean log-feature vectors is the only distance used here. It compares first moments through one pooled covariance, inverted through the pseudo-inverse because the nine features are exactly

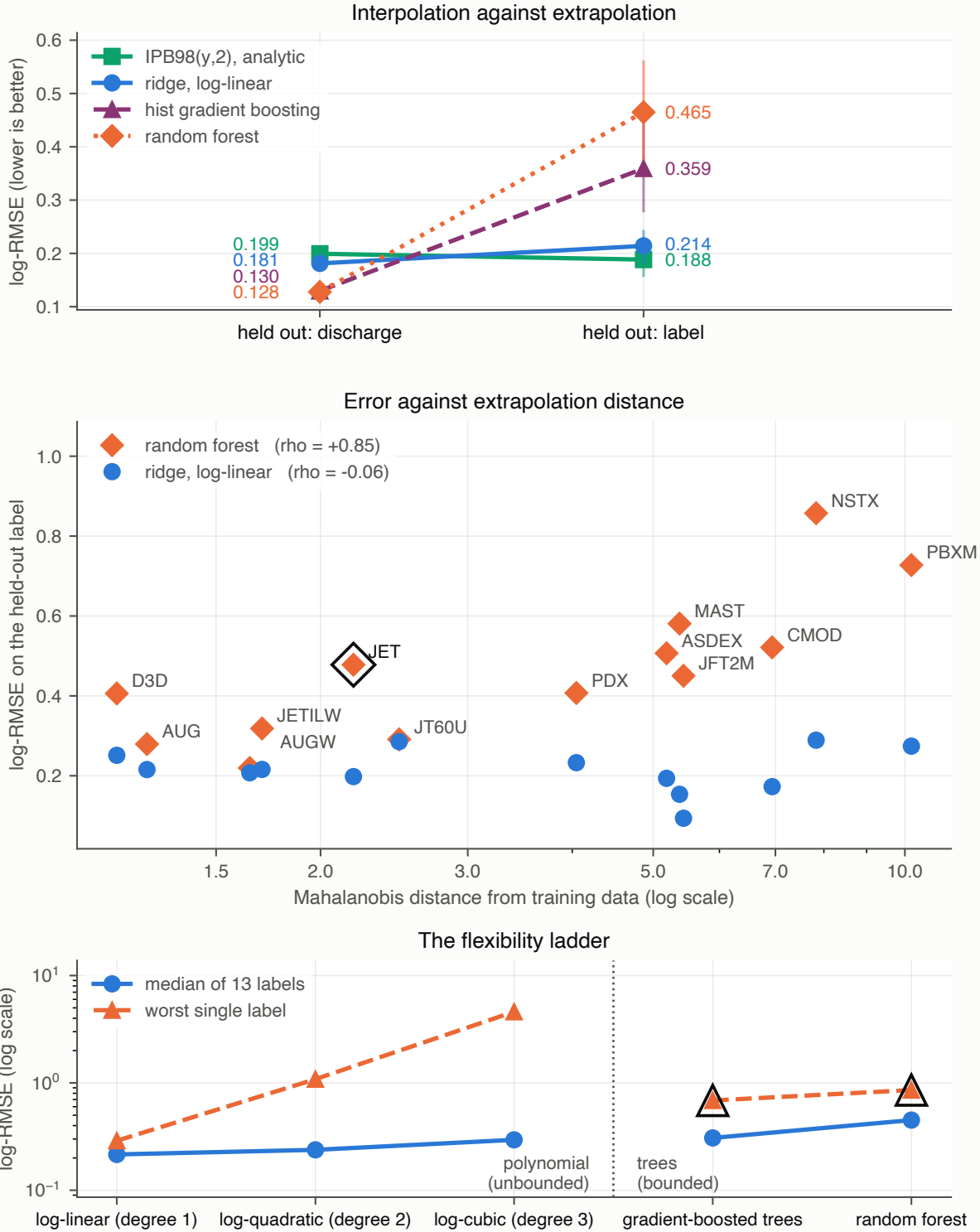


FIG. 1. **Interpolation against extrapolation.** Top: each model under both splits, with 95% percentile-bootstrap intervals over the 13 held-out labels. Middle: per-label error against distance from the training data. Bottom: the flexibility ladder.

TABLE I. The headline comparison: the same models and the same nine engineering features, scored under each split. LOLO is leave-one-label-out over the 13 eligible labels. Rows are ordered by cross-validation score, and the six models above the rule are refitted inside every fold; the two tree ensembles are tied under cross-validation at the resolution of Sec. VIII. IPB98(y,2) is a historical reference, not refitted within folds. Section IIIB matches the row population and the aggregation between the columns and repeats the comparison over physical devices.

Model	CV	LOLO	Ratio
random forest	0.128	0.465	3.6×
hist gradient boosting	0.130	0.359	2.8×
ridge, log-cubic	0.148	0.849	5.7×
ridge, log-quadratic	0.158	0.300	1.9×
ridge, log-linear	0.181	0.214	1.2×
mean baseline	0.869	0.994	1.1×
<i>Historical reference, not fold-fitted</i>			
IPB98(y,2), analytic	0.199	0.188	1.0×

collinear (Sec. S14), so it is blind to differences in the shape of an operating envelope at a fixed mean. The diagnosis carries over to dimensionless coordinates (Sec. S12). Section S19 gives the permutation tests and the stability of ρ when each label is dropped, and Sec. S8 the per-label scores.

b. A ceiling that binds only once. A random forest averages leaf values, and every leaf value is a mean of training targets, so it cannot predict outside $[\min y_{\text{train}}, \max y_{\text{train}}]$. That bound binds on one held-out label, JET (circled in the middle panel of Fig. 1), where 48% of the rows exceed the largest confinement time in the training fold, and across the size cut of Sec. IV, where **34%** of held-out rows do. It does not explain the pattern. The four labels the forest loses by the widest margins are NSTX (+0.568), PBX-M (+0.453), MAST (+0.427) and JFT-2M (+0.357), and on each its ceiling stands about three natural logs above anything the held-out machine reaches. JET, the one fold where the ceiling is in the way, is seventh of the thirteen, at +0.280. The forest fails hardest where the bound is furthest from binding, so the failure happens inside the training range: its predictions stop tracking the features once the input leaves the support it was fitted on. The gradient booster has

TABLE II. The random forest and the power law under both row populations, both aggregations, and both definitions of a machine. The forest leads under cross-validation and trails the power law under leave-one-out in all eight combinations.

Split	pooled over rows		each unit equally	
	forest	ridge	forest	ridge
CV, all 18 labels	0.128	0.181	0.149	0.238
CV, 13 scored labels	0.123	0.177	0.124	0.187
LOLO, 13 labels	0.410	0.214	0.465	0.214
LODO, 11 devices	0.551	0.200	0.527	0.211

no such guarantee, but on every split scored here its predictions stayed below the training maximum (Sec. S19).

c. Flexibility costs the tail. A polynomial ladder in the log features (degrees 1 to 3) is flexible and does extrapolate. Degree 2 is not meaningfully worse than degree 1 on a typical label, at a median of 0.238 against 0.216, but its worst label scores 1.083 against degree 1’s 0.289, and degree 3’s worst scores 4.601 (Fig. 1, bottom, where the two ensembles are circled). Every flexible model scored so far is therefore either bounded, like the two ensembles, or divergent, like the cubic. Nothing yet separates “flexible models fail out of distribution” from “models that stop trending outside their data fail out of distribution”. Section VI B builds the model that does.

B. Controls

Three things other than the held-out unit differ between the columns of Table I. Cross-validation scores all 6228 rows while leave-one-out scores only the 13 labels large enough to hold out. It pools residuals over rows, where JET and ASDEX Upgrade dominate, while leave-one-out averages per label. And the wall variants are separate labels. Table II varies all three. The forest leads under every cross-validation cell and trails under every leave-one-out cell, and the published law’s score barely moves, from 0.188 per label to 0.184 per device.

Two objections could explain the reversal without any appeal to long-range behaviour, and Table III collects the controls for both.

TABLE III. Controls for the two objections, each scored with the same models and splits as Table I. CV is the grouped cross-validation log-RMSE of the forest and the power law. *Lost* counts the held-out labels (of 13) and physical devices (of 11) on which the forest is worse than the power law, and the gaps are the mean paired differences in log-RMSE, positive where the forest is worse. The stored-energy and dimensionless arms are scored on the 6214 rows that carry a stored energy. Secs. S18 and S11 give the construction of each arm.

Arm	CV	lost		gap	
	forest / power law	labels	devices	label	device
confinement time, nine features	0.128 / 0.181	13	11	+0.251	+0.315
stored energy as the target	0.123 / 0.193	9	9	+0.220	+0.348
loss power deleted	0.198 / 0.315	10	9	+0.051	+0.070
dimensionless groups	0.229 / 0.322	11	9	+0.074	+0.130
within-device features only	0.157 / 0.315	6	5	−0.033	−0.018

a. The loss power is the target’s denominator. The thermal confinement time is the stored energy over the loss power, and the loss power is one of the nine features. A log-linear model represents that identity exactly, outside the training range of P as well as inside it, where a tree cannot. Regressing the stored energy instead keeps P as an ordinary predictor and removes only the identity. On the 6214 rows that carry a stored energy, the forest still wins cross-validation, 0.123 against 0.193. Under leave-one-out it loses on 9 of 13 labels and 9 of 11 devices, at mean gaps of +0.220 and +0.348, and across the size cut it scores **1.081** against the power law’s 0.289. Deleting P outright is the harsher test, because it removes a real predictor as well, and the forest then loses on 10 of 13 labels and 9 of 11 devices. The identity therefore carries the unanimity of the count, and under either control neither exact test reaches $p = 0.05$, but it does not carry the effect.

b. Do the features name the machine? Four of the nine features barely move once a device is fixed. Major radius carries **0.1%** of its log variance within devices, and a random forest classifier recovers the device from the nine features at **99.9%** under the discharge-grouped split. A tree can therefore fit a per-device offset under cross-validation that is worthless on a machine it has not seen. Deleting the near-constant features removes the reversal, and the forest’s interpolation margin grows to 50.3%, but that control also deletes

the size dependence the power law extrapolates through, so it cannot tell the two readings apart. Dimensionless coordinates come closer, and fitting this database in them is established practice^{2,27}. In $(\rho^*, \beta, \nu^*, q, \epsilon, \kappa, M)$ the size, field and density dependence is kept while no coordinate is a device label: ρ^* moves 26.6% within a device, against major radius's 0.1%. Regressing $B\tau_E$ on those seven groups, the forest wins cross-validation by **28.9%** and loses on **11 of 13** labels and **9 of 11** devices. The device gap falls from +0.315 to +0.130: part of the engineering-space gap is device identity and part is not. This arm carries a confound of its own. The temperature the groups need is recovered from the stored energy as $T \sim W_{\text{th}}/(nV)$ rather than measured, so the groups are algebraic functions of the target, neither model is blind here in the way it is in engineering space, and the coordinates cannot place a machine that does not exist yet. ϵ and κ also still move only 5% and 6% within a device. It is a different experiment with a different confound, not a stricter version of the same one (Sec. S18). This database does not cleanly separate device identity from size physics.

c. Other choices. None of the remaining choices a reader could suspect carries the reversal (Sec. S11). The power law wins every scored label at eligibility thresholds of 20, 30 and 50 rows, and 14 of 15 at 10. Over ten shuffled fold assignments the forest's cross-validation margin stays between 27.0% and 28.5%. A nested hyperparameter search improves the forest's held-out label score from 0.465 to 0.403, or 0.376 with the inner folds grouped by machine, against the power law's 0.214. Dropping the spherical tokamaks leaves the forest worse on all 11 remaining labels. A mixed model with a random intercept per device, the estimator the clustered-validation literature pairs with this split, scores **0.829** per device against the power law's 0.212 (Sec. S10). On the full DB5.2.3 revision, on 5358 H-mode rows STD5 does not contain and 3860 ohmic and L-mode rows, a tree ensemble loses to the refitted power law on an unseen label in **16 of 24** label-model pairs and **9 of 10** (Sec. S2).

C. A database with no device in common

The stellarator-heliotron replication of Sec. S17 of the supplementary material was locked before its data were downloaded, and the outcome was to be reported under a predeclared rule whichever way it fell. On the standard set, 1719 rows over 7 devices scored at the same 30-row threshold as above, the random forest is better than the log-linear power law

under cross-validation, 0.209 against 0.288, and worse on **7 of 7** eligible held-out devices, 0.814 against 0.442, a paired gap of +0.373 [+0.106, +0.797]. The booster is worse on 7 of 7. Under the rule fixed in advance that is a reproduction of the inversion, on devices the tokamak database does not contain, and the Gaussian process again scores lowest under both splits and again does not reverse. One thing differs: the power law’s error rises with distance there, at $\rho = +0.57$ against the forest’s $\rho = +0.64$, where on the tokamak database it is flat. The supplement traces that to the device-by-device constant ISS04 itself needs.

IV. A SIZE CUT MATCHED TO ITER

Holding out JET still leaves the rest of the database spanning much of its size range, so leave-one-out extrapolates in identity while interpolating in size. ITER’s major radius is 6.2 m against 3.40 m for the largest row here, a factor of 1.82, and that factor is available inside the database. Training on the 14 smallest database labels (up to DIII-D, 1.865 m, 3498 rows) and predicting the 4 largest (up to JT-60U, 3.40 m, 2730 rows) asks for a ratio of 1.823 against ITER’s 1.824. The cut is chosen by that ratio, not by eye.

Table IV sets the cut beside the two easier splits. The power law scores 0.278 there, the analytic law 0.194 and a constant predictor 1.459. The random forest scores 0.938 and the gradient booster 1.072, so the two models that lead the conventional comparison land nearer a constant than the power law. The ordering is the same on each of the three individually scored held-out labels. The two newer published laws fitted to this database revision, ITPA20 and ITPA20-IL², score **0.165** and **0.158** across the cut, each evaluated on the areal elongation it is written on (Sec. S13). Both were fitted to selections that include the machines above the cut, so they are references here, not blind competitors. Dropping the spherical tokamaks moves the random forest only from 0.938 to 0.936. A bounded correction learned on the power law’s residuals and built for this cut takes the power law from 0.278 to 0.206 (Sec. S9).

The cut has one sharp limit. Of its 2730 held-out rows, 1762 are JET and 866 JET-ILW, so 96% come from one physical device and it is close to a single-device holdout at a size ratio. Why the size dependence in this database is weaker than IPB98(y,2)’s is an active question^{28,29}, and across this cut $\log \rho^*$ and $\log \nu^*$ separate as well as size does (Sec. S12). The cut is therefore a joint step in size, normalised gyroradius and collisionality, and no

TABLE IV. Three questions of increasing difficulty, scored in log-RMSE. The *size cut* column is the ITER-size-matched cut. The two models that win the first column land closer to the constant baseline than to the power law in the third.

Model	discharge CV	LOLO	size cut
IPB98(y,2)	0.199	0.188	0.194
ridge, log-linear	0.181	0.214	0.278
random forest	0.128	0.465	0.938
hist grad. boosting	0.130	0.359	1.072
mean baseline	0.869	0.994	1.459

claim rests on it alone. The claim that rests on many machines is the leave-one-device-out result of Sec. III.

V. CALIBRATED INTERVALS THAT MISS WITHOUT WIDENING

For a next-step device the point error is not the deliverable; the interval is. Split-conformal intervals³⁰ are calibrated here on absolute log residuals, holding back 25% of the *discharges* in each training fold, and the half-width is the finite-sample conformal rank of those scores at $\alpha = 0.10$ (Sec. S7 gives the construction). Kardaun³¹ built confidence and prediction intervals for ITER’s confinement time from the predecessor database by a parametric route. The question here is narrower: whether an interval calibrated the usual way still covers once the held-out unit is a device.

Under grouped cross-validation every model covers within a point of 90% (Table V). On an unseen label the random forest’s 90% interval covers 35% of rows, and across the ITER-size-matched cut 3%. The gradient booster covers 8 of the 2730 rows there. A calibration bug would give the same number, so it is worth checking by hand. The booster’s half-width across the cut is 0.221 in logs, and its per-label errors above the cut run from 0.66 to 1.13, three to five times the half-width, so its intervals have no chance of containing the held-out values. Every coverage in Table V pools rows. With each held-out label weighted equally instead, the forest’s 35% becomes 26% and the booster’s 45% becomes 41%. Above the cut, neither tree ensemble’s interval covers more than 6% of any label. Cross-validation over the

TABLE V. Empirical coverage of nominal 90% split-conformal intervals, and the multiplicative half-width under leave-one-out.

Model	CV	LOLO	ITER cut	width
IPB98(y,2)	90%	89%	88%	1.40×
ridge, log-linear	90%	83%	70%	1.33×
bounded hybrid	90%	64%	76%	1.26×
hist grad. boosting	91%	45%	0%	1.23×
random forest	91%	35%	3%	1.23×

13 scored labels stays within two points of nominal for every model, so the collapse does not depend on the aggregation.

The widths barely move. No model’s interval changes width by more than 1.5% between the two arms, and it could not be otherwise: the half-width is a rank statistic of the calibration scores, which are drawn the same way in both arms, so the test distribution has no route to it. What is measured is the coverage. The fixed width is the practical problem, because the interval carries no sign that it has stopped covering. The assumption that fails is exchangeability, which the weighted-conformal literature can restore only by knowing the shift³², and calibrated uncertainty degrading under dataset shift is a general finding rather than a fusion one³³. Coverage falls with distance for the trees, at $\rho = -0.77$ for the forest and -0.54 for the booster, and not for the power law, at $+0.27$ (Fig. 2). Rows arrive in correlated clusters of slices from one discharge, so every coverage here is an empirical rate rather than a guaranteed one.

If the collapse is about exchangeability, calibrating on held-out labels instead of held-out discharges should repair it, and only as far as that unit reaches. Both halves hold. On a held-out label, coverage returns to within two points of nominal for every model, the random forest going from 35% to 88%. Across the ITER-size-matched cut none of the schemes tested restores it, because every calibration unit is smaller than the test unit, and scaling the interval by distance lifts the forest only from 3% to 40%. Calibrated by device and pooled over rows the repair is less complete still: the booster reaches 77%, or 93% with each device weighted equally, and the trees buy their coverage with multiplicative half-widths of 2.3 to 2.8, where the linear models stay near 1.4. The constrained power law of Sec. VIA is

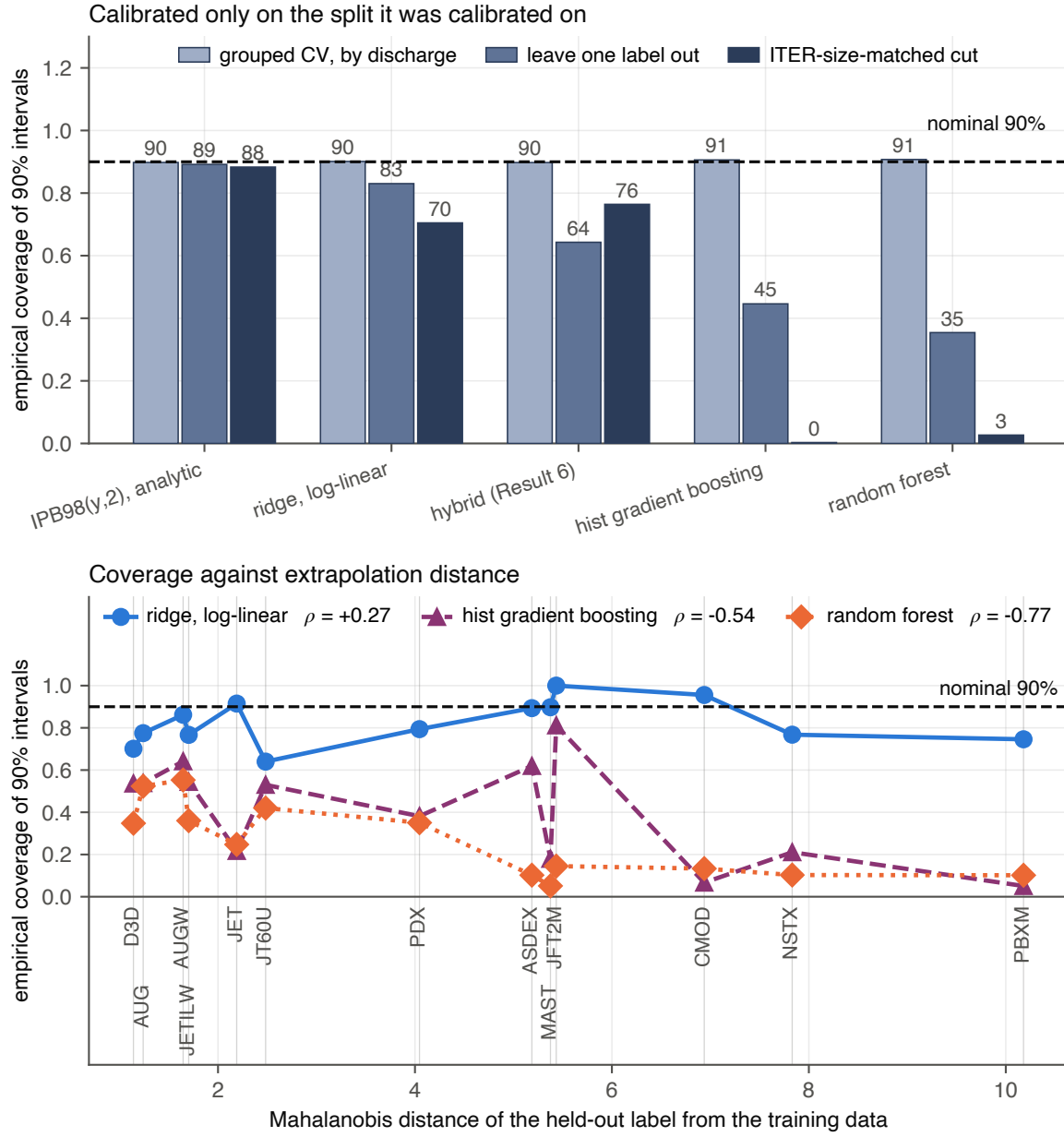


FIG. 2. **The interval collapse.** Top: empirical coverage of nominal 90% split-conformal intervals for each model under the three splits, against the nominal line. Every model is calibrated under grouped cross-validation, and only the linear and constrained fits stay near nominal once the held-out unit becomes a whole label and then a size range. Bottom: per-label coverage against the Mahalanobis distance of the label from the training data. Interval widths behind both panels move by under 1.5% between the splits.

the only refitted model whose plain split-conformal coverage stays near nominal at the size cut, at 91% (Sec. S1).

VI. WHAT EXTRAPOLATES: LONG-RANGE STRUCTURE

A. Connor–Taylor constraints on the exponents

Every model to this point learns its functional form from the data. Connor and Taylor³⁴ showed that requiring a confinement scaling law to be expressible in the dimensionless plasma variables imposes *linear equality constraints on its exponents*, and dimensionless-parameter scaling experiments work in the same transformation family^{35,36}. A physics assumption is then a matrix C , and the fit becomes

$$\min_b \|Xb - y\|^2 \quad \text{subject to} \quad Cb = d, \quad (2)$$

which the Karush–Kuhn–Tucker system solves in closed form. The constraints are derived in code from the definitions of the groups, written as exponents on (length, field, temperature, density),

$$\rho^* \sim T^{1/2} B^{-1} L^{-1}, \quad \beta \sim n T B^{-2}, \quad \nu^* \sim n L T^{-2}. \quad (3)$$

A law expressible in whichever groups a model holds fixed must be invariant under every scale transformation that leaves them unchanged, and each such transformation is one row of C . Holding ρ^* , β and ν^* fixed (Kadomtsev³⁷) gives one row, holding ρ^* and β (collisionless) two, and holding ρ^* alone (electrostatic) three (Sec. S15).

IPB98(y,2) is itself an instance: the high- β Kadomtsev constraint was enforced when it was fitted, with the field exponent held at 0.15^{1,2}. Its exponents lie within 0.00096 of the Kadomtsev surface derived here, which checks the derivation. It may also be part of why a law fitted in 1998 to a predecessor database transfers as well as it does. On a held-out label IPB98(y,2) scores 0.188 against 0.214 for the unconstrained refit on the same rows and features, and at the size cut 0.194 against 0.278. The newer ITPA20 laws sit further from every surface and transfer better still, so closeness to a surface does not order laws fitted to different populations (Sec. S15). What the hierarchy tests is narrower: a refit on these rows in which the constraint is the only difference.

The Kadomtsev rung is *prospective*. Kadomtsev specified it in 1975, and it was enforced when IPB98(y,2) was fitted, so nothing about it was selected here. It is free: at 0.1808 in

TABLE VI. Fitting under the Connor–Taylor hierarchy, ordered by score at the ITER-size-matched cut. Between the first and last rows only the constraint differs: the same features, the same solver, no hyperparameter. The Kadomtsev row is prospective, fixed by the literature before this analysis; the collisionless row was selected on the scores in this table. LODO is the held-out score over the 11 physical devices. *Worse on* counts the devices on which a row scores above the unconstrained fit; the last row, two solvers for one model, calibrates that column.

Model	in sample	CV	LOLO	LODO	worse on	ITER cut
power law, collisionless	0.1818	0.182	0.206	0.203	5/11	0.183
IPB98(y,2), analytic*	n/a	0.199	0.188	0.184	4/11	0.194
bounded hybrid (Sec. S9)	n/a	0.151	0.246	0.246	5/11	0.206
power law, electrostatic	0.2245	0.225	0.241	0.226	8/11	0.237
power law, Kadomtsev	0.1808	0.181	0.210	0.211	7/11	0.254
power law, unconstrained	0.1808	0.181	0.214	0.212	8/11 [†]	0.279

*Historical reference, fitted to the DB2.8 predecessor of this database and not refitted within these folds.

[†]The unconstrained fit scored against the ridge spelling of the same power law: the calibration for the column, not a result.

sample it matches the unconstrained fit to four decimals, because the data already satisfy it. It still beats the unconstrained fit at every one of the 15 cuts in the size sweep, including all 8 well-powered ones, those that train on at least 1000 rows. At six training machines the unconstrained fit scores 0.971 and the Kadomtsev fit 0.274. A constraint stops a fit wandering along the weak direction of the design when the training half is small or shifted. Where it is neither, on the leave-one-out arms, the rung gains 0.004 by label and 0.001 by device, inside the resolution of Sec. VIII, and at gaps that small the per-device counts in Table VI are coin tosses (Sec. S15).

The collisionless rung is the stronger assumption and the better score. It costs 0.001 in sample, wins at 8 of 8 well-powered cuts, and at the ITER-size-matched cut scores **0.183**, the lowest error of any model fitted here without the held-out rows. It was chosen after the three rungs had been scored at that cut, so 0.183 is the best of the surfaces examined and not a prediction. ITER is expected to be less collisional than the machines in the database, which points at this rung for a reason outside the data, but that does not make the result

prospective. Two published laws score lower at the cut, ITPA20-IL at 0.158 and ITPA20 at 0.165, and both had seen the machines above it. One rung further, the electrostatic law degrades to 0.237. Cross-validation prefers none of the constraints, so it cannot select the rung that transfers best, and supplying the same physics as a shrinkage prior rather than a constraint is worth much less (Sec. S15). Figure 3 shows the trade.

B. Flexibility or long-range saturation?

Flexibility is how much structure a model can learn beyond a power law. *Saturation* is whether its predictions stop tracking the features once the input leaves the training support. The forest is provably confined to its training range, the booster never left it, and a squared-exponential Gaussian process reverts to its prior mean. The ensembles are flexible and saturating, ridge is neither, and the cubic is flexible and non-saturating but divergent. A Gaussian process³⁸ with a linear-plus-RBF kernel, a linear term added to a squared-exponential (radial-basis-function, RBF) term, is flexible, non-saturating and well behaved at long range. The tool is used in this field to fit plasma profiles³⁹; the kernel is chosen here for its behaviour far outside the data. Three kernels fitted over the same features with the same optimiser (Sec. S6) order as saturation predicts. The squared-exponential kernel alone scores 1.948 at the ITER-size-matched cut, worse than a constant, with an error tracking distance at $\rho = +0.65$. Adding a linear term gives the lowest cross-validated score in this study, 0.112 against the forest’s 0.128, with 0.191 at the cut and an error flat against distance, at $\rho = -0.01$.

A linear term enlarges the function class as well as changing the asymptote, so that ladder suggests the mechanism without isolating it. The ablation of Table VII isolates it. Write the model as a mean function plus a residual process,

$$\log \tau = m(x) + g_{\text{RBF}}(x), \quad (4)$$

and fit g identically in every arm, with the same covariance family, tuning subsample, seed and optimiser; only m changes. With a constant mean the model reproduces the squared-exponential rung to every digit computed, which is the control. A fitted power-law mean moves the cut from 1.948 to 0.278, a factor of 7.0, and 0.278 is plain ridge’s score there to six decimals: far from the data the residual process has decayed and the model returns

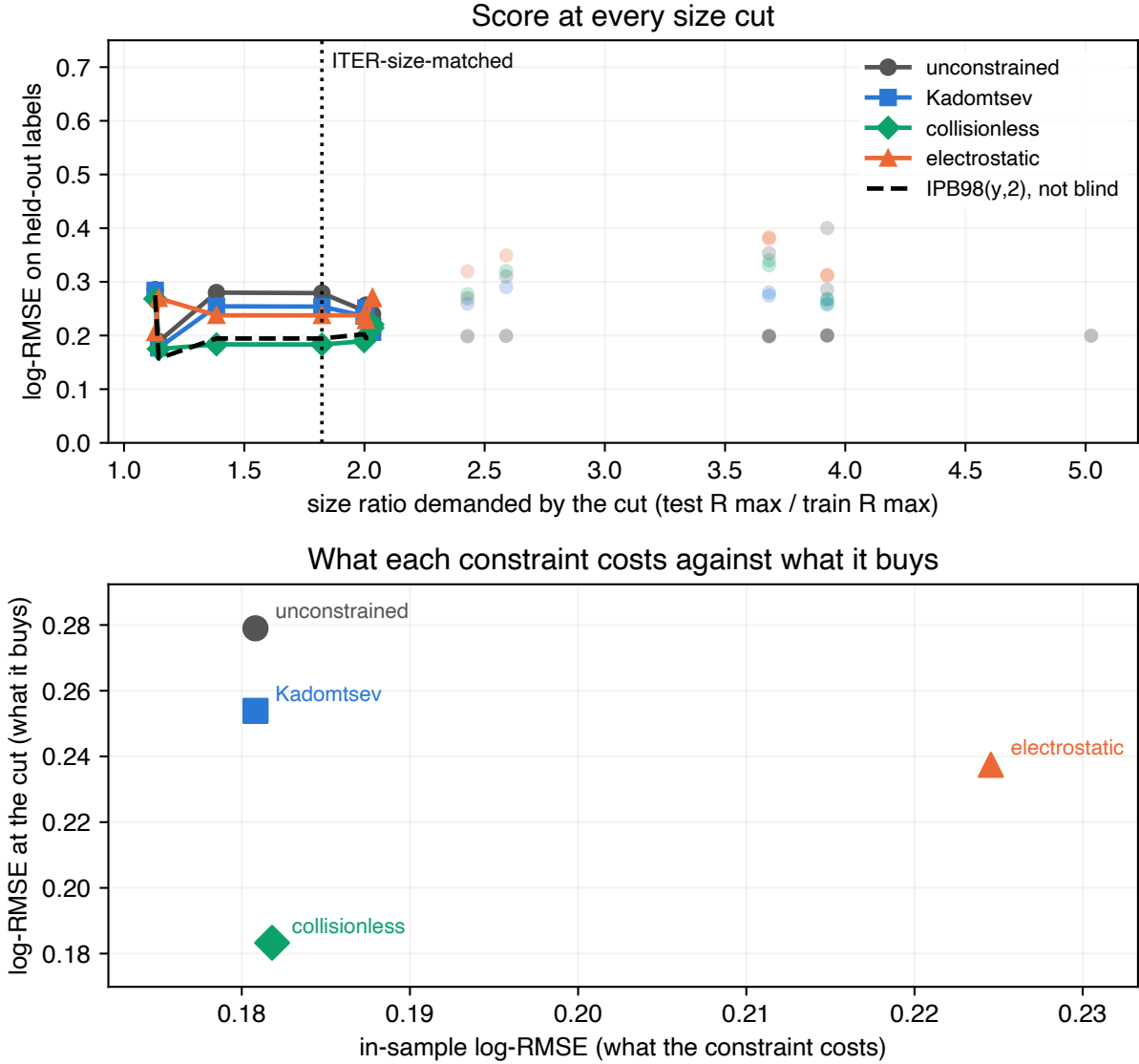


FIG. 3. **Cost against benefit across the size sweep.** Top: the score at every size cut for the constrained and unconstrained fits, with underpowered cuts drawn faint and excluded from claims. Bottom: what each constraint costs in sample against what it buys at the ITER-size-matched cut. The hierarchy has an optimum in its middle, at the collisionless rung, which was itself selected on these scores.

its mean. A third arm keeps that from being a tautology. Its mean also trends, with IPB98(y,2)'s published exponents, fitted to nothing here. Swapping it in moves the cut to **0.186** and the held-out label score from 0.212 to 0.189. Having a trend is necessary, and which trend it is decides the answer.

TABLE VII. One RBF covariance family and fitting procedure, three mean functions. The constant-mean row reproduces the squared-exponential rung of Sec. S6 to six decimals, so the decomposition is the same model rewritten. Only the first row saturates. The other two both keep trending, and they do not agree. LODO repeats the held-out score over the 11 physical devices.

$m(x)$ in Eq. (4)	CV	held-out	LODO	ITER cut
constant	0.142	0.541	0.693	1.948
fitted power law	0.115	0.212	0.212	0.278
IPB98(y,2), unfitted	0.112	0.189	0.186	0.186

Across the model families tested, then, extrapolation failure tracks long-range saturation rather than flexibility alone. The practical rule concerns saturation: avoid models whose predictions stop tracking the features once the input leaves the training data, whether or not they are flexible. The process does not displace the constrained power law: across the cut it scores 0.191 against 0.183, a tie at the resolution of Sec. VIII, and held out one physical device at a time it scores 0.219 against the power law’s 0.211, worse on 5 of the 11. What it adds is an interval that needs no calibration. At the ITER-size-matched cut its native 90% posterior interval covers 92.5% of rows, where split-conformal intervals give the random forest 3% and the gradient booster none. The squared-exponential rung is the instructive failure. It is the one model that does widen out of distribution, with a half-width four times any other, and it still covers only 16.7%, because it has reverted to a prior mean nowhere near the answer. A wider interval does not help when it is centred in the wrong place.

VII. DISCUSSION

Conventional within-database evaluation does not measure what a scaling law is used for. Cross-validation grouped by discharge, shot or time slice holds out rows from machines that stay in the training set, so it measures interpolation within known devices. On this database the two comparisons do not merely differ in size; they order the candidate models in opposite directions, and a gain reported under the first does not establish a gain under the second. Among the three primary models the convention picks the worst extrapolator. The failure depends on the candidate set. When the set includes a model whose predictions keep

trending, such as the linear-plus-RBF process, cross-validation selects it, and on a held-out device it stays level with the power law. Adding that model removes the failure for this candidate set. It does not show that discharge-grouped cross-validation selects reliably for unseen devices in general. Tree ensembles are fitted to this database¹⁵; whether Bayesian¹⁴ or feature-aligned¹⁶ networks saturate is a question about those architectures that only this split would settle, and none of them is scored here.

The stellarator replication says the inversion is not a property of one compilation, of the standard set’s selection criteria or of the machines the ITPA database happens to hold, since it appears, under a rule written before the data were seen, on a database with no device in common. It does not establish a rate across confinement databases or across architectures: it is one more database and the same three models. The ITER-size-matched stress test, the interval collapse and the mechanism experiment rest on the tokamak database alone. A further external replication, on the IAEA/IEA CICLOP multi-machine database^{40,41}, was specified and locked in the archived repository before its data were requested; the database had not been opened to the author by submission, so no result from it is reported, and the lock stays in force for whatever the data show. Two replications outside fusion, on Kleiber’s exponent⁴² and on the allometry of the BAAD compilation⁴³, reproduce the extrapolation failure (Secs. S4 and S5). They also do what these nine features cannot: they include cases where the flexible model does not win the interpolation split, and the failure appears there too, which is the case that gives no warning at all.

A. What this changes in practice

The consequence is mainly a reporting convention. For any confinement model offered as an alternative to a power law:

1. Report leave-one-device-out beside cross-validation. It costs one refit per device, and where the two disagree, as here by a factor of 3.6, the disagreement is the result.
2. Report how far the target operating point sits from the training distribution, the one quantity known in advance for a machine that does not exist. What separates the models is the contrast, $\rho = +0.85$ for the random forest against $\rho = -0.06$ for the power law, since even the mean baseline correlates with distance.

3. Say whether the model’s predictions keep tracking the features outside the training support, and whether they can leave the range of the training targets at all; a random forest provably cannot. Both are known before anything is fitted.
4. Do not carry a calibrated interval across a device boundary without saying so, and do not read its width as evidence. Here 90% nominal delivered 35% on an unseen label and 3% across the size cut, at a width that moved by under 1.5%.

Only the first needs a refit.

The positive results point the same way. A Connor–Taylor constraint and a linear kernel term are unrelated operations, and both supply long-range structure. For practice, the cheaper model was also the stronger one: a power law refitted on the same rows costs one least-squares solve and is a stronger baseline out of distribution than any tree ensemble scored here. Where a flexible model is wanted, the evidence favours one that keeps trending. Where a physics constraint is available, imposing it is close to free.

What the models predict for SPARC, JT-60SA and ITER is locked in Sec. S3 of the supplementary material, with a lock date of 31 August 2026 and a content digest over the input rows. None of the three has a measured H-mode thermal confinement time at the tabulated operating point, so the table is a record of predictions, not a validation. On JT-60SA, inside the database’s size range, all five models agree to within 15%. On ITER, $1.82\times$ beyond it, they disagree by a factor of 8.3, and the random forest cannot close its gap, because its prediction cannot exceed the largest training target, 1.321 s. At ITER’s $Q = 10$ inductive point the constrained power law predicts 2.837 s and the unconstrained refit 2.858 s, against IPB98(y,2)’s 3.591 s, so both fits made here sit **21%** below the reference law at the operating point it was used to size, in the direction the newer published laws also point. That estimate comes from a rung selected on the size-cut scores, and it is conditional on the operating point of the 1999 basis¹ as carried forward in 2007³, which ITER has since rebaselined⁴⁴. SPARC sits further from the training data than ITER, at a Mahalanobis distance of 6.9 against 4.7, because its field is 2.1 times the largest in the database: size is not the only direction in which a next-step device leaves the data behind.

VIII. LIMITATIONS

Resolution. Two models whose per-label scores differ by less than about 0.01 in log-RMSE are tied here; that is roughly the bootstrap spread of one label’s score and what a fold assignment moves the cross-validated margin by. The orderings this paper claims, such as the forest’s 0.465 against 0.214 and the size cut’s 0.938 against 0.278, are an order of magnitude clear of it.

One machine above the cut. The ITER-size-matched cut reproduces ITER’s size ratio and nothing else: ITER is a burning plasma dominated by alpha heating at a gain no row here approaches, so clearing this bar is necessary and not sufficient. Of the 2730 held-out rows, 2628 are one physical device. The interval collapse, the bounded correction, the constrained fit’s 0.183, the Gaussian process’s 0.191 and the mean-function ablation are all scored across it, so all five are one-machine evidence about a large jump. A second large device at comparable weight is what the comparison needs, and the database does not contain one.

Population and weighting. JET and ASDEX Upgrade with their variants supply 77% of rows, and every fit weights rows equally. Weighting each label equally costs the power law more than the forest and takes the win count to 12 of 13 (Sec. S11). Only 13 of the 18 labels are scored, and the five left out are the smallest.

Selection. Across the splits, published laws, fitted models, constraint rungs, kernels, mean functions, size cuts, thresholds, partitions, feature sets and targets, this paper reports of order forty out-of-distribution comparisons, with no correction for multiplicity; every p -value among them is an exact sign test over at most thirteen dependent units. The descriptive gaps that carry the paper are far clear of its resolution, but its constructive results come from a search, and exactly one of them is prospective: the Kadomtsev rung. The collisionless rung, the damping of the bounded correction and the interval scheme of the locked forecast were each chosen after a held-out score had been seen, and should be read as exploratory.

Errors in variables. Every power-law fit treats the engineering parameters as exact, which the confinement-scaling literature has long objected to^{27,31,45}. Orthogonal distance regression on this design runs away along its weak direction, so no errors-in-variables fit is offered (Sec. S16).

Model classes. The mixed model has a single random intercept and no random slopes (Sec. S10). The Gaussian-process intervals are equal-tailed posterior intervals on $\log \tau$ that include the fitted noise term, the closest analogue of a conformal interval but not the same construction, and the kernels are tuned on a seeded subsample of each training fold (Sec. S6). The Gaussian-process result is for one kernel family, and a process with a linear part constrained as in Sec. VIA, which would combine the two things that work here, is untested. The two replications outside fusion carry their own limits, given in Secs. S4 and S5. Of the three locked forecasts, only JT-60SA’s can be checked in the near term, and only once it runs an H-mode campaign.

IX. CONCLUSION

On real confinement data, the model that wins cross-validation by 29% over a log-linear power law refitted in the same folds loses to that power law on all 13 eligible held-out database labels and on all 11 eligible physical devices. Controls for the loss power in the target and for features that identify the device remove the unanimity of that count but not the effect, although the device-identity control is not blind to the target. The inversion reproduces on a stellarator-heliotron database under rules fixed before its data were seen. At the size step that separates this database from ITER, the two tree ensembles land closer to a constant predictor than to the power law, and their nominal 90% intervals cover 3% and 0% of held-out rows at unchanged width.

What improved extrapolation here was long-range structure: a Connor–Taylor constraint on the exponents, whose Kadomtsev rung costs nothing in sample, and a Gaussian process whose mean keeps trending. Across the models tested, the failure belongs to models whose predictions stop tracking the features once the input leaves the training data, not to flexible models as such. A scaling law is asked for a number at a point no row occupies, so the validation used to choose one should hold out whole devices, since a device is what the deployment will not have seen.

SUPPLEMENTARY MATERIAL

See the supplementary material for the construction, tables and caveats behind every result this paper summarises (Secs. S1 to S20), including the per-label scores, the model and split specification, the replications, and the provenance of every input.

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AUTHOR DECLARATIONS

Conflict of interest. The author has no conflicts to disclose.

Author contributions. Liam Salcedo: Conceptualization (lead); Data curation (lead); Formal analysis (lead); Investigation (lead); Methodology (lead); Project administration (lead); Software (lead); Validation (lead); Visualization (lead); Writing – original draft (lead); Writing – review & editing (lead).

DATA AVAILABILITY

The data that support the findings of this study are openly available. Five third-party datasets underlie this work, and none is redistributed here: each is fetched from its public source by a script in the repository and verified against a pinned SHA-256 on load, and Sec. S20 of the supplementary material lists the fingerprints.

- **ITPA Global H-mode Confinement Database, standard analysis set STD5 (DB5.2.3).** The basis of every fusion result. Openly available from the OSF deposit *International Global H-Mode Confinement Database*²⁴ and described in its reference paper². File `hdb5_std5.csv`.
- **The full DB5.2.3 revision.** Used for the replication on rows STD5 excludes (Sec. S2) and for the stored-energy, dimensionless and elongation controls, which take `WTH` and `KAPPA` from it. A separate download from the same OSF project (<https://osf.io/download/zhwa3/>). File `hdb5_db523.csv`.
- **Mammalian basal metabolic rate.** The Figshare deposit of Capellini, Venditti and Barton⁴⁶. Used in the supplementary material. File `allometry_bmr.txt`.
- **Biomass And Allometry Database (BAAD) v1.0.1.** Released CC0; the data paper is cited above⁴³ and the archive analysed here is release v1.0.1⁴⁷. Used in the supplementary material. File `baad_data.zip`.
- **International Stellarator-Heliotron Confinement Database, version 26.** The external replication of Sec. S17 of the supplementary material. Maintained by the IEA stellarator-heliotron collaboration, described with the ISS04 scaling in its reference paper²², and served from an open directory at IPP Greifswald (<https://ishpdb.ipp-hgw.mpg.de/ISS/>). The analysed file is `ISHCDB_26.txt`, 3311757 bytes, pinned at SHA-256 `3a9f857b768e....`

The code, analysis scripts and test suite that regenerate every number and figure in this paper and its supplementary material are openly available under the MIT licence at <https://github.com/lsalcedo100/FusionFlux>, and the version used here is archived with a persistent identifier⁴⁸ at doi:10.5281/zenodo.22971392 (v0.4.15; the DOI for all versions is 10.5281/zenodo.22215142). Every number and figure was produced at commit

8ab13e44f048bba7e1b3c84e59d96d6b824b3cd2, and every script verifies the pinned data fingerprints as it loads and refuses to run if they disagree. **One hundred and twenty-seven** of the numbers printed in this paper, and **one hundred and eighty-three** across it and its supplementary material, are tied by test to the field of the generated file they came from, so a rerun that moved a value, or an edit that changed one, fails the test suite (Sec. S20).

REFERENCES

- ¹ITER Physics Expert Group on Confinement and Transport, ITER Physics Expert Group on Confinement Modelling and Database and ITER Physics Basis Editors, “Chapter 2: Plasma confinement and transport,” *Nucl. Fusion* **39**, 2175 (1999). doi:10.1088/0029-5515/39/12/302
- ²G. Verdoolaege *et al.*, “The updated ITPA global H-mode confinement database: description and analysis,” *Nucl. Fusion* **61**, 076006 (2021). doi:10.1088/1741-4326/abdb91
- ³E. J. Doyle *et al.*, “Chapter 2: Plasma confinement and transport,” *Progress in the ITER Physics Basis*, *Nucl. Fusion* **47**, S18 (2007). doi:10.1088/0029-5515/47/6/S02
- ⁴A.J. Creely *et al.*, “Overview of the SPARC tokamak,” *J. Plasma Phys.* **86**, 865860502 (2020). doi:10.1017/S0022377820001257
- ⁵J. Citrin *et al.*, “Real-time capable first principle based modelling of tokamak turbulent transport,” *Nucl. Fusion* **55**, 092001 (2015). doi:10.1088/0029-5515/55/9/092001
- ⁶O. Meneghini *et al.*, “Self-consistent core-pedestal transport simulations with neural network accelerated models,” *Nucl. Fusion* **57**, 086034 (2017). doi:10.1088/1741-4326/aa7776
- ⁷P. Rodriguez-Fernandez *et al.*, “Predictions of core plasma performance for the SPARC tokamak,” *J. Plasma Phys.* **86**, 865860503 (2020). doi:10.1017/S0022377820001075
- ⁸L. Allen and C. M. Bishop, “Neural network approach to energy confinement scaling in tokamaks,” *Plasma Phys. Control. Fusion* **34**, 1291 (1992). doi:10.1088/0741-3335/34/7/008
- ⁹J. Kates-Harbeck, A. Svyatkovskiy and W. Tang, “Predicting disruptive instabilities in controlled fusion plasmas through deep learning,” *Nature* **568**, 526 (2019). doi:10.1038/s41586-019-1116-4
- ¹⁰C. Rea, R.S. Granetz, K. Montes, R.A. Tinguely, N. Eidietis, J.M. Hanson and B. Sammuli,

- “Disruption prediction investigations using machine learning tools on DIII-D and Alcator C-Mod,” *Plasma Phys. Control. Fusion* **60**, 084004 (2018). doi:10.1088/1361-6587/aac7fe
- ¹¹J.X. Zhu *et al.*, “Hybrid deep-learning architecture for general disruption prediction across multiple tokamaks,” *Nucl. Fusion* **61**, 026007 (2021). doi:10.1088/1741-4326/abc664
- ¹²J. Degraeve *et al.*, “Magnetic control of tokamak plasmas through deep reinforcement learning,” *Nature* **602**, 414 (2022). doi:10.1038/s41586-021-04301-9
- ¹³A. Murari, M. Peluso, M. Gelfusa and I. Lupelli, “Symbolic regression via genetic programming for data driven derivation of confinement scaling laws without any assumption on their mathematical form,” *Plasma Phys. Control. Fusion* **57**, 014008 (2015). doi:10.1088/0741-3335/57/1/014008
- ¹⁴E. Gao *et al.*, “Bayesian neural networks for predicting tokamak energy confinement time with uncertainty quantification,” *Nucl. Fusion* **65**, 084001 (2025). doi:10.1088/1741-4326/ade8fd
- ¹⁵X. Xiong, K. Zhang, Y. Gao, L. Zhang, X. Chen and S. Liu, “Bayesian optimization based machine learning for predicting tokamak energy confinement time with quantification of feature importance,” *Fusion Eng. Des.* **227**, 115740 (2026). doi:10.1016/j.fusengdes.2026.115740
- ¹⁶H. Nam and J. Seo, “Machine learning based energy confinement time extrapolation via multi-tokamak database,” *Fusion Eng. Des.* **221**, 115369 (2025). doi:10.1016/j.fusengdes.2025.115369
- ¹⁷Z. Wang *et al.*, “Power-law-anchored residual learning for H-mode energy confinement time in tokamaks: interpolation and parameter-defined extrapolation,” arXiv:2608.20684 (2026). doi:10.48550/arXiv.2608.20684
- ¹⁸D.R. Roberts *et al.*, “Cross-validation strategies for data with temporal, spatial, hierarchical, or phylogenetic structure,” *Ecography* **40**, 913 (2017). doi:10.1111/ecog.02881
- ¹⁹B. Meredig *et al.*, “Can machine learning identify the next high-temperature superconductor? Examining extrapolation performance for materials discovery,” *Mol. Syst. Des. Eng.* **3**, 819 (2018). doi:10.1039/C8ME00012C
- ²⁰C. Sutton *et al.*, “Identifying domains of applicability of machine learning models for materials science,” *Nat. Commun.* **11**, 4428 (2020). doi:10.1038/s41467-020-17112-9
- ²¹J. Abbate *et al.*, “Large-database cross-verification and validation of tokamak transport models using baselines for comparison,” *Phys. Plasmas* **31**, 042506 (2024).

doi:10.1063/5.0190908

- ²²H. Yamada *et al.*, “Characterization of energy confinement in net-current free plasmas using the extended International Stellarator Database,” *Nucl. Fusion* **45**, 1684 (2005). doi:10.1088/0029-5515/45/12/024
- ²³J. Butt, G. Verdoolaege, S. M. Kaye and E. Kolemen, “Revisiting confinement scalings and fusion performance with a perspective optimized for extrapolation,” arXiv:2604.22728 (2026). doi:10.48550/arXiv.2604.22728
- ²⁴ITPA Confinement Database and Modelling Topical Group, *International Global H-Mode Confinement Database*, version 5.2.3 [dataset], Open Science Framework (2021). <https://osf.io/drwcq>
- ²⁵L. Breiman, “Random forests,” *Machine Learning* **45**, 5 (2001). doi:10.1023/A:1010933404324
- ²⁶M.J. Leyland *et al.*, “The H-mode pedestal structure and its role on confinement in JET with a carbon and metal wall,” *Nucl. Fusion* **55**, 013019 (2015). doi:10.1088/0029-5515/55/1/013019
- ²⁷J. G. Cordey *et al.*, “Scaling of the energy confinement time with β and collisionality approaching ITER conditions,” *Nucl. Fusion* **45**, 1078 (2005). doi:10.1088/0029-5515/45/9/007
- ²⁸J. Hall, P. Zhang and G. Verdoolaege, “Confinement scaling with machine size in the updated ITPA global H-mode energy confinement database,” in *Proc. 49th EPS Conf. on Plasma Physics*, Bordeaux, Europhys. Conf. Abstr. **47A**, P1.024 (2023).
- ²⁹J. Hall, P. Zhang, Y. Zhang, C. Angioni and G. Verdoolaege, “Confinement scaling with machine size in the updated ITPA global H-mode energy confinement database,” poster P2.1097, *52nd EPS Conf. on Plasma Physics*, Edinburgh (2026). Conference presentation; not published in refereed form at the time of writing.
- ³⁰V. Vovk, A. Gammerman and G. Shafer, *Algorithmic Learning in a Random World*, Springer (2005). doi:10.1007/b106715
- ³¹O. J. W. F. Kardaun, “Interval estimation of global H-mode energy confinement in ITER,” *Plasma Phys. Control. Fusion* **41**, 429 (1999). doi:10.1088/0741-3335/41/4/001
- ³²R.J. Tibshirani, R. Foygel Barber, E.J. Candès and A. Ramdas, “Conformal prediction under covariate shift,” *Adv. Neural Inf. Process. Syst.* **32**, 2526 (2019).
- ³³Y. Ovadia *et al.*, “Can you trust your model’s uncertainty? Evaluating predictive uncer-

- tainty under dataset shift,” *Adv. Neural Inf. Process. Syst.* **32**, 13969 (2019).
- ³⁴J.W. Connor and J.B. Taylor, “Scaling laws for plasma confinement,” *Nucl. Fusion* **17**, 1047 (1977). doi:10.1088/0029-5515/17/5/015
- ³⁵C. C. Petty, “Sizing up plasmas using dimensionless parameters,” *Phys. Plasmas* **15**, 080501 (2008). doi:10.1063/1.2961043
- ³⁶T. C. Luce, C. C. Petty and J. G. Cordey, “Application of dimensionless parameter scaling techniques to the design and interpretation of magnetic fusion experiments,” *Plasma Phys. Control. Fusion* **50**, 043001 (2008). doi:10.1088/0741-3335/50/4/043001
- ³⁷B.B. Kadomtsev, “Tokamaks and dimensional analysis,” *Sov. J. Plasma Phys.* **1**, 295 (1975).
- ³⁸C.E. Rasmussen and C.K.I. Williams, *Gaussian Processes for Machine Learning*, MIT Press (2006). doi:10.7551/mitpress/3206.001.0001
- ³⁹M.A. Chilenski, M. Greenwald, Y. Marzouk, N.T. Howard, A.E. White, J.E. Rice and J.R. Walk, “Improved profile fitting and quantification of uncertainty in experimental measurements of impurity transport coefficients using Gaussian process regression,” *Nucl. Fusion* **55**, 023012 (2015). doi:10.1088/0029-5515/55/2/023012
- ⁴⁰X. Litaudon *et al.*, “Long plasma duration operation analyses with an international multi-machine (tokamaks and stellarators) database,” *Nucl. Fusion* **64**, 015001 (2024). doi:10.1088/1741-4326/ad0606
- ⁴¹X. Litaudon *et al.*, “Investigating long-duration plasma operation with the international multi-machine CICLOP database,” *Nucl. Fusion* **66**, 095001 (2026). doi:10.1088/1741-4326/ae89cc
- ⁴²M. Kleiber, “Body size and metabolism,” *Hilgardia* **6**, 315 (1932). doi:10.3733/hilg.v06n11p315
- ⁴³D.S. Falster *et al.*, “BAAD: a Biomass And Allometry Database for woody plants,” *Ecology* **96**, 1445 (2015). Released CC0. doi:10.1890/14-1889.1
- ⁴⁴A. Loarte *et al.*, “The new ITER baseline, research plan and open R&D issues,” *Plasma Phys. Control. Fusion* **67**, 065023 (2025). doi:10.1088/1361-6587/add9c9
- ⁴⁵G. Verdoolaege, A. Shabbir and G. Hornung, “Robust analysis of trends in noisy tokamak confinement data using geodesic least squares regression,” *Rev. Sci. Instrum.* **87**, 11D423 (2016). doi:10.1063/1.4960069
- ⁴⁶I. Capellini, C. Venditti and R.A. Barton, “Supplement 1: data on basal

metabolic rate with experimental animal body mass” [dataset], Figshare (2016). doi:10.6084/m9.figshare.3549807.v1

⁴⁷D.S. Falster *et al.*, *BAAD: a Biomass And Allometry Database for woody plants* [dataset], Figshare (2015). doi:10.6084/m9.figshare.c.3307692. Release v1.0.1, <https://github.com/dfalster/baad/releases/tag/v1.0.1>

⁴⁸L. Salcedo, *FusionFlux: extrapolation, long-range saturation and physics constraints in confinement scaling*, version v0.4.15, Zenodo (2026). doi:10.5281/zenodo.22971392